

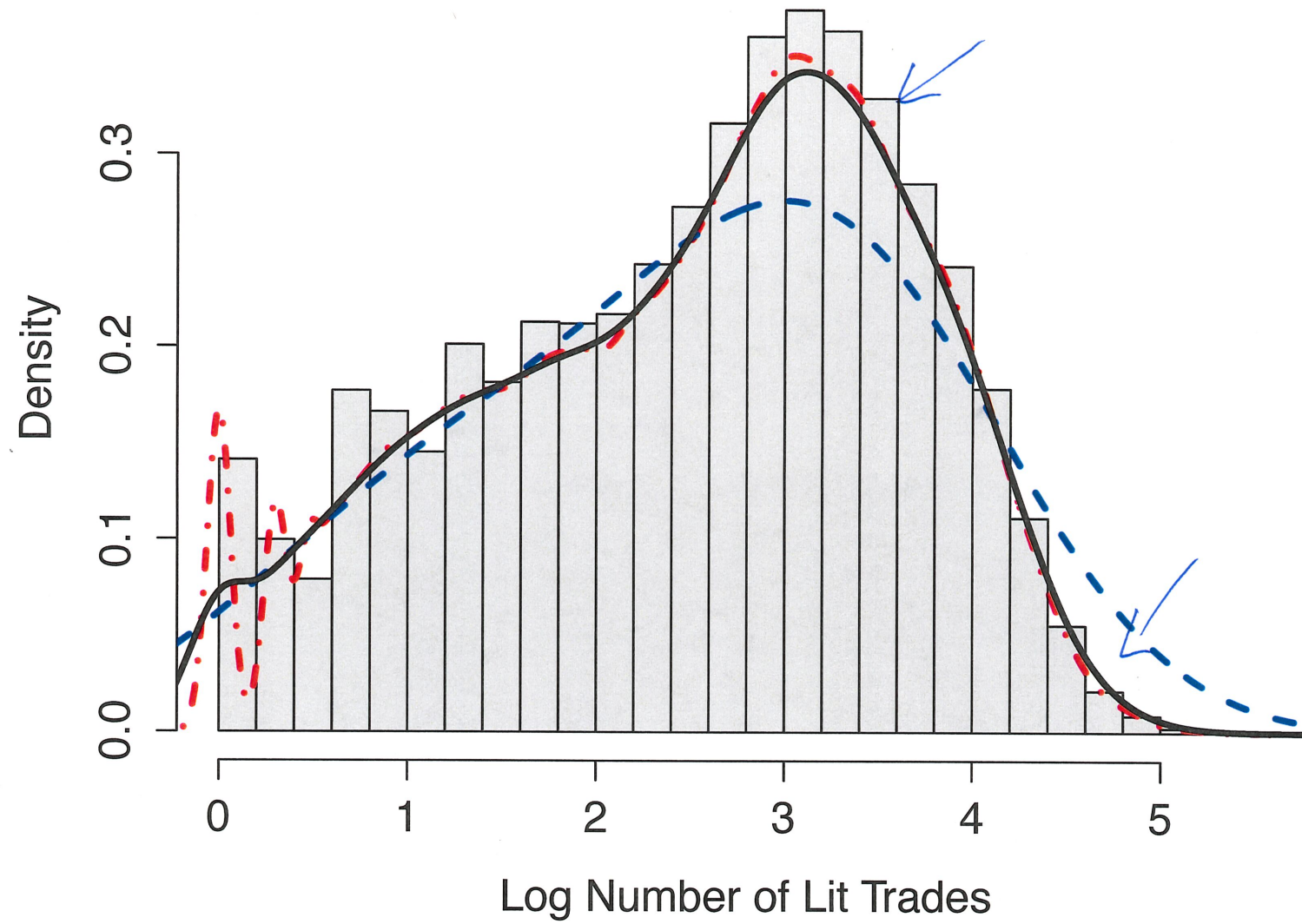
1 Part 1: Introduction to Modelling

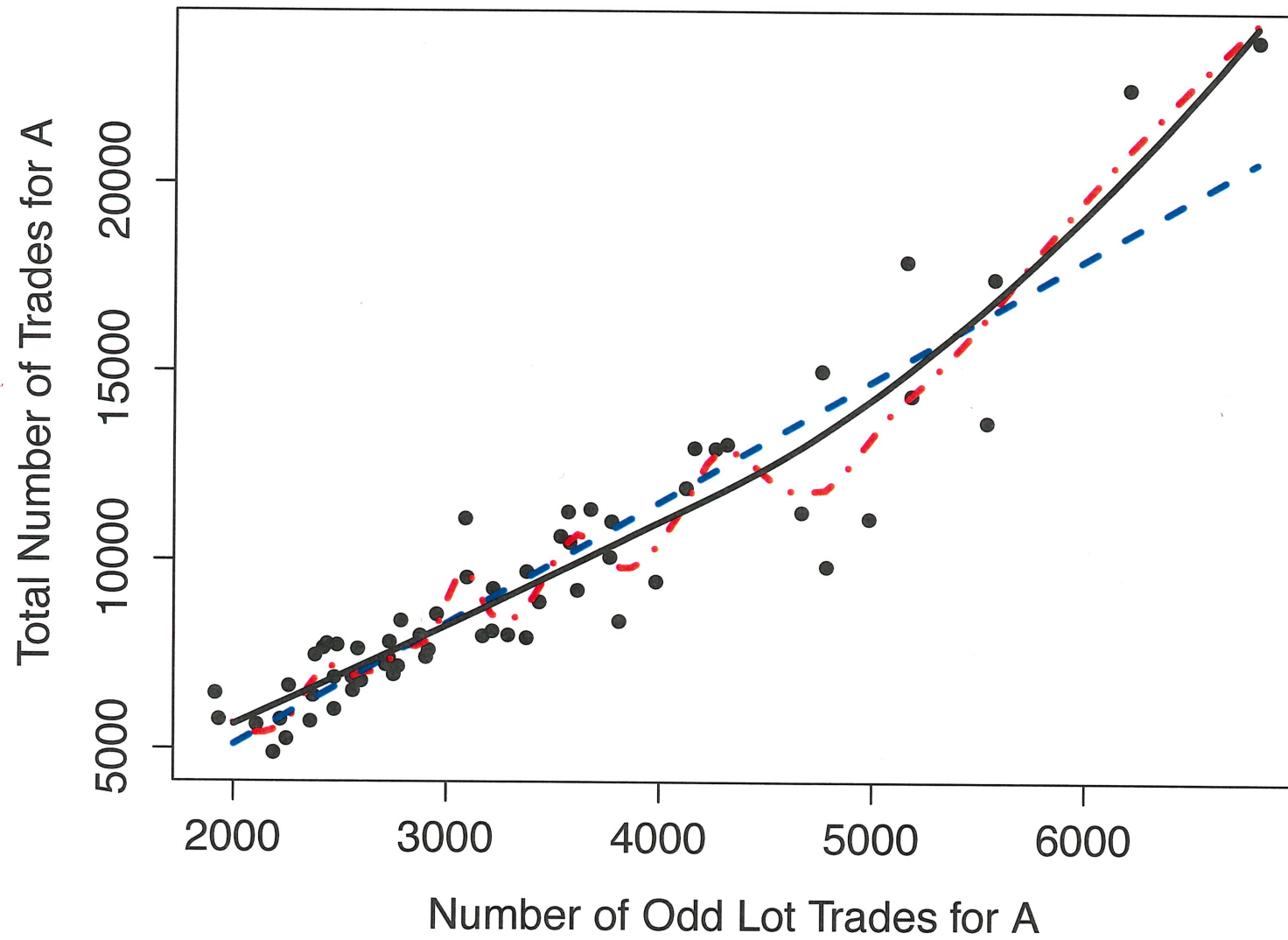
2 A fundamental step in many data analyses is to **model** raw, observable
3 data, i.e. to replace the complex processes that generate these data with a
4 mathematically and computationally simpler fiction.

5 You have likely heard the expression, "**All models are wrong; some are**
6 **useful.**"¹ That is a good mentality to take towards the modelling process:
7 We do not believe that the data were actually generated by, for example,
8 drawing from a normal distribution. Instead, it **may** be the case that the
9 normal provides a sufficiently accurate approximation to that process.

¹For a history of this saying, see https://en.wikipedia.org/wiki/All_models_are_wrong.

- 1 A key heuristic in the statistical modelling process is that of **smoothness**.
- 2 We believe that most relationships, and most distributions, are fundamen-
- 3 tally **smooth**, i.e., we don't try to fit to every little fluctuation that our sam-
- 4 ple may possess.
- 5 Of course, this idea can be taken too far, and **oversmoothing** can lead to
- 6 missing important features that are present in data.
- 7 (The market structure data on the following two slides are from the SEC, Quarter 1 of 2017,
- 8 https://www.sec.gov/opa/data/market-structure/marketstructuredownloadshhtml-by_security.html)





- 1 **Exercise:** Explain how the three proposed models on the previous slides
2 seem to perform, in regards to the appropriate amount of smoothing of the
3 histogram and scatter plot.

4 In each case, the dashed (blue) line is the "smoothest."

5 In fact, in both cases, they are oversmoothing, missing
6 seemingly important features.

7 The dash-dotted line (red) is the least smooth.

8 It is fitting to small fluctuations ("artifacts") in
9 the sample. This is overfitting.

10 The black (solid) line is a compromise between the
11 two.
12

1 **Exercise:** Explain why overfitting is such a significant concern.

2 "overfitting" $\leftrightarrow$ "undersmoothing"

3
4 A key objective is to extend from the training set
5 to the larger population from which that set
6 was drawn.

7
8 An overfit model is fit "artifacts" that are present
9 only in the training set. There could be large
10 errors in your predictions, and you would be
11 overconfident in those predictions because error
12 bars (standard errors, CIs) are overly optimistic

1 In many situations, there are one or more unknown **population quantities**
2 that are of interest to estimate. For example,

- 3 • The probability that the return exceeds c tomorrow
- 4 • The “beta” for some stock over the past year.
- 5 • The volatility for some stock over the past three months.
- 6 • The mean of some distribution of interest.
- 7 • The variance of some distribution of interest.
- 8 • The λ parameter from some Exponential distribution.



9 The model serves as a means to estimating such a quantity, since we imag-
10 ine that the true model, if known, would provide the true value of the
11 unknown. An estimated model provides an estimate of the unknown.

- 1 From a statistical standpoint, we say that a probability model is **incom-**
- 2 **pletely specified**. I.e., a particular model is assumed, but there are still
- 3 unknown components, often real-valued **parameters**.
- 4 We use available data (a **sample**) to estimate these parameters.
- 5 The quantities of interest such as those listed above are either the parame-
- 6 ters themselves, or some function of those parameters.
- 7 **Example:** In a certain situation, an individual is comfortable assuming that
- 8 $X_1, X_2, \dots, X_n$ are **independent and identically distributed (i.i.d.)** with the
- 9 Normal distribution with mean μ and variance σ^2 . These parameters are
- 10 unknown, and must be estimated from data.

- 1 **Exercise:** Suppose someone wants to estimate the probability that tomorrow's return will be less than c . Describe steps as to how a model could be
2 constructed and the quantity of interest estimated.
3

4 ① Obtain daily historical data for this stock

5 ② Choose a model for the dist., e.g. $\text{Normal}(\mu, \sigma^2)$
6

7 ③ Use the data to estimate μ, σ^2 , - call these $\hat{\mu}, \hat{\sigma}^2$ ← estimator
8

9 ④ The prob. of interest is some function of
10 μ, σ^2 , i.e. $g(\mu, \sigma^2)$. So, estimate using $g(\hat{\mu}, \hat{\sigma}^2)$
11

12 ⑤ Attach a measure of uncertainty to my estimate,
ie. report a CI for $g(\mu, \sigma^2)$

1 Terminology

2 The **parameters** are the constants that distinguish probability models in the
3 same “family.” Examples of “families” include “normal,” “exponential,”
4 and so forth.

5 A **statistic** is a function of the RVs being used to model the observable data.
6 Examples include the sample mean, the sample variance, etc. A statistic is
7 a **random variable**.

$$\bar{X} = g(X_1, X_2, \dots, X_n) \quad S^2 \quad \bar{X}(\omega) = g(X_1(\omega), X_2(\omega), \dots, X_n(\omega))$$

8 An **estimator** is a statistic which is used to estimate an unknown parameter
9 or an unknown function of a parameter. It is also a **random variable**.

10 The “hat notation” is standard for denoting an estimator: $\hat{\beta}$ is an estimator
11 for β , for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its
12 estimator.

- 1 **Exercise:** Explain the sense in which an estimator is appropriately thought
2 of as a random variable.

3 An estimator is a function of the RVs used to model
4 the observable data.

5 Let $\hat{\theta}$ be an estimator for θ .

6
$$\hat{\theta}(\omega) = g(X_1(\omega), X_2(\omega), \dots, X_n(\omega)) \quad \omega \in \Omega$$

7 ie. $\hat{\theta}$ is a function on Ω
8

9
$$E(\hat{\theta}), \quad v(\hat{\theta}), \quad P(\hat{\theta} \leq a)$$

10

1 Performance of Estimators

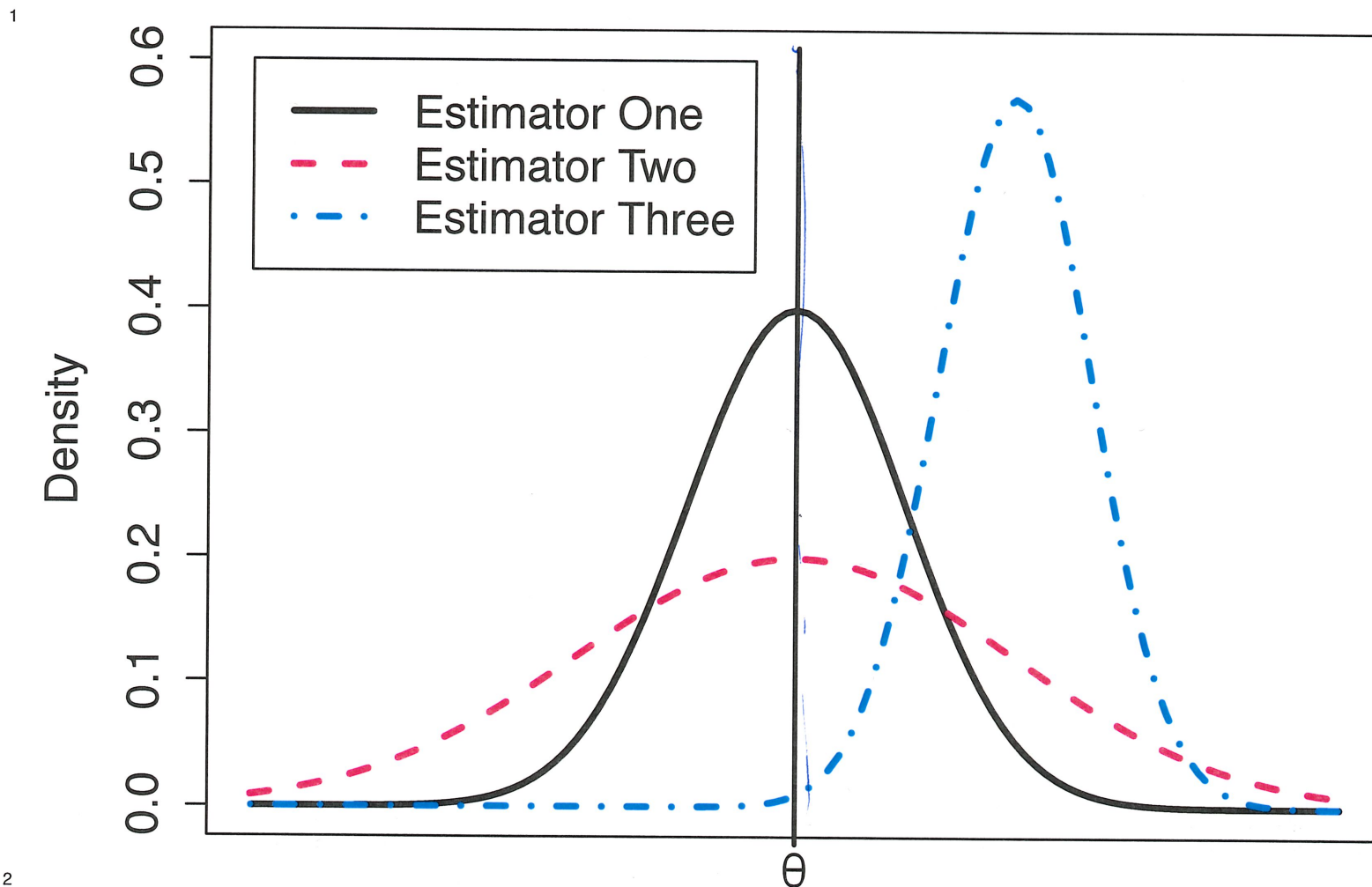
2 Our objective is to estimate unknowns, i.e., to construct estimators. For
3 any situation, there are many choices of estimator, so we require methods
4 to assess the performance of an estimator.

5 The “performance” is closely tied to the distribution of the estimator. The
6 distributions we learned in Probability (plus some more) are useful for this
7 purpose.

8 An ideal estimator $\hat{\theta}$ for θ would have two properties:

- 9 1. The estimator should be **accurate**, i.e., its distribution would be cen-
10 tered on the true value θ .
- 11 2. The estimator should be **precise**, i.e., its distribution has a small spread
12 (small variance)

- Example:** The figure on the following page compares the distributions of three hypothetical estimators for a parameter θ , call them $\hat{\theta}_1$, $\hat{\theta}_2$, and $\hat{\theta}_3$.
- We note that $E(\hat{\theta}_1) = \theta$, so it is optimally accurate. Also, $E(\hat{\theta}_2) = \theta$.
- But, $V(\hat{\theta}_1) < V(\hat{\theta}_2)$, so $\hat{\theta}_1$ is more precise than $\hat{\theta}_2$.
- So, we would prefer $\hat{\theta}_1$ to $\hat{\theta}_2$.
- Also, $V(\hat{\theta}_3) < V(\hat{\theta}_1)$, but $\hat{\theta}_3$ has terrible accuracy.



- 1 An estimator $\hat{\theta}$ is said to be **an unbiased estimator for θ** if $E(\hat{\theta}) = \theta$.
- 2 The variability in an estimator (its precision) is naturally quantified by its
- 3 variance, $V(\hat{\theta})$.
- 4 The **standard error (SE) of $\hat{\theta}$** is the square root of its variance. The SE is of-
- 5 ten reported along with an estimate in order to give a sense of the amount
- 6 of error one should anticipate in that estimate.
- 7 An estimator $\hat{\theta}$ is said to be the **minimum variance unbiased estimator for**
- 8 **θ (MVUE)** if $\hat{\theta}$ is unbiased for θ and for any other unbiased estimator $\hat{\theta}'$,
- 9 $V(\hat{\theta}) \leq V(\hat{\theta}')$.

- 1 **Exercise:** Suppose that X has the binomial(n, p) distribution. It is com-
2 mon to estimate p using $\hat{p} = X/n$, the sample proportion. Is this estimator
3 unbiased? ^{for p .} What is its variance and standard error?

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} (np) = p$$

Yes, $\hat{p}$ is an unbiased estimator for p .

$$V(\hat{p}) = V\left(\frac{X}{n}\right) = \frac{1}{n^2} V(X) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

$$SE \text{ of } \hat{p} = \sqrt{V(\hat{p})} = \sqrt{\frac{p(1-p)}{n}}$$

1 There is often a “tradeoff” between bias and variance (accuracy and preci-
 2 sion). This is called the **bias-variance tradeoff**.

3 In some cases, allowing some bias leads to reduced variance.

4 This tradeoff is made formal via the **mean squared error (MSE)**. For an
 5 estimator $\hat{\theta}$ for θ ,

$$6 \quad \text{MSE}_{\theta}(\hat{\theta}) = E\left((\hat{\theta} - \theta)^2\right)$$

7 i.e., it is the expected squared error in estimating θ when using $\hat{\theta}$.

8 One can show that

$$9 \quad \text{MSE}_{\theta}(\hat{\theta}) = V(\hat{\theta}) + \text{bias}_{\theta}^2(\hat{\theta})$$

10 where

$$11 \quad \text{bias}_{\theta}(\hat{\theta}) = \underline{E(\hat{\theta})} - \theta$$

E.g. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$, $E(S^2) = \sigma^2$ under general conditions

$\hat{\sigma}^2 = \frac{1}{n+1} \sum (X_i - \bar{X})^2$, when $X_1, X_2, \dots, X_n$ are iid $N(\mu, \sigma^2)$, minimizes
 MSE $\hat{\sigma}^2 = \left(\frac{n-1}{n+1}\right) S^2$ $V(\hat{\sigma}^2) = \left(\frac{n-1}{n+1}\right)^2 V(S^2) < V(S^2)$

- 1 **Exercise:** How does the bias/variance tradeoff relate to our initial discus-
2 sion of smoothness?

3 Smooth models have lower variance, i.e. don't fluctuate
4 as much when a new sample is drawn

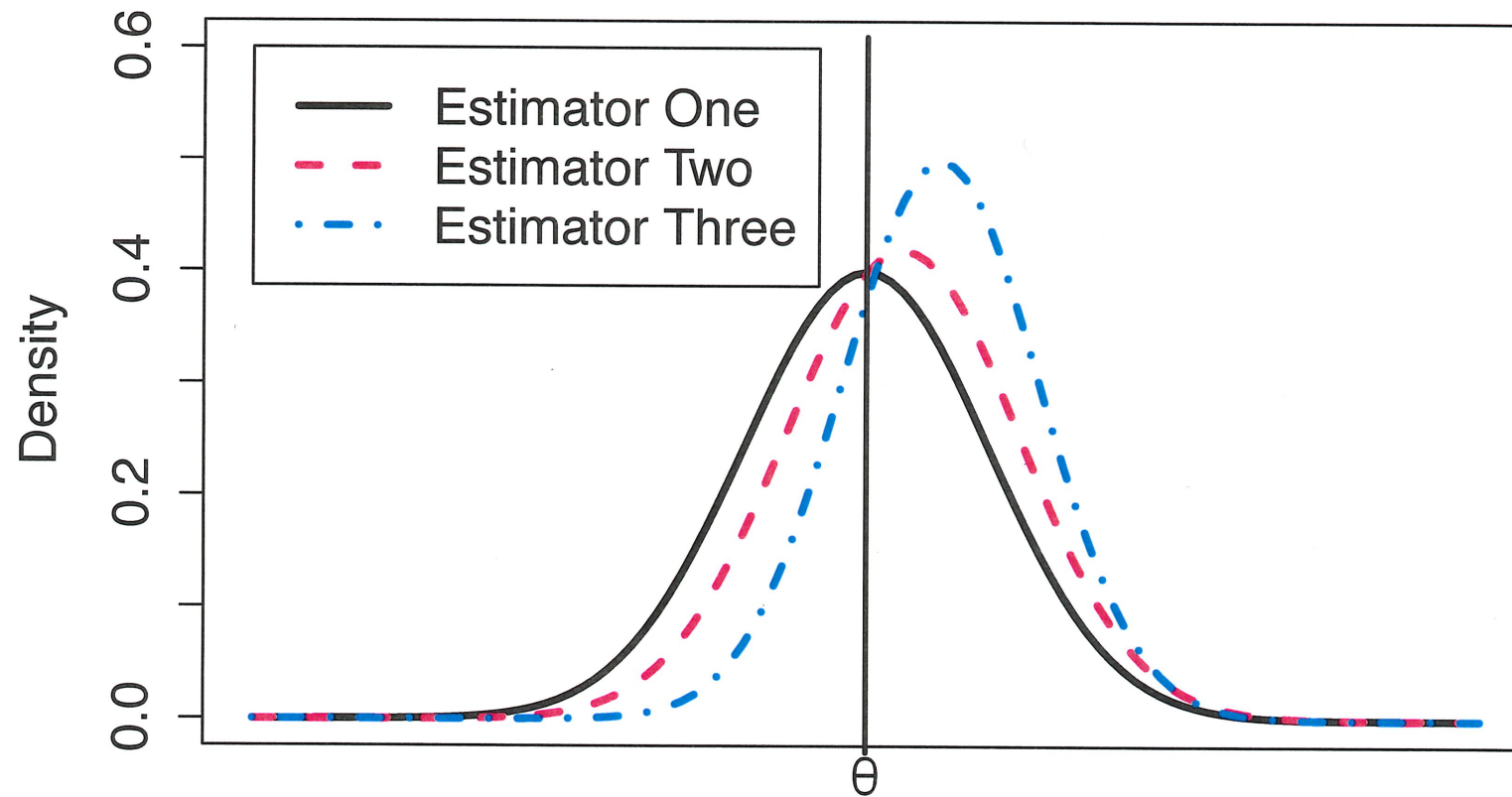
5
6 Less smooth models have larger variance, as they
7 try to fit to artifacts in the new sample

8
9 Smooth models have larger bias. The more we
10 smooth, the more we miss real features, i.e. bias.

11 Less smooth models have less bias.

- 1 In the example below, all three estimators have the same MSE.

2



3

1 Important Cases

2 There are a few estimation situations that worth special attention.

3 **Case 1:** The first we have already explored, that of estimating a population
4 proportion. We found that the sample proportion is unbiased and has a
5 standard error of

6
$$\sqrt{\frac{p(1-p)}{n}}$$

7 Of course, we do not know the value of p , so typically it is replaced by its
8 best estimate and the approximate SE is reported as

9
$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- 1 **Exercise:** We have derived the mean and variance of the sample propor-
2 tion, but what is its distribution?

3 $\hat{p}$ is approximately Normal by CLT

4 $\hat{p} = \frac{\sum X_i}{n}$ where $X_i = \begin{cases} 1, & \text{when trial } i \text{ is "success"} \\ 0, & \text{" " " " "failure"} \end{cases}$

5 $X_1, X_2, \dots, X_n$ iid Bernoulli(p)

6
7 The approx. improves as n grows. Also better
8 when $p \approx 0.5$.

9
10 Rule of thumb: Approx. acceptable if $np \geq 5$ and $n(1-p) \geq 5$

1 **Case 2:** Suppose that $X_1, X_2, \dots, X_n$ are random variables and $E(X_i) = \mu$
2 and $V(X_i) = \sigma^2$ for all i . Further assume that these random variables
3 are uncorrelated (meaning that every pair of random variables is uncorre-
4 lated).

5 Then, the standard estimator for μ is the sample mean, $\bar{X}$, and the standard
6 estimator for σ^2 is the sample variance,

7
$$s^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1)$$

8 Note that the case where $X_1, X_2, \dots, X_n$ are i.i.d. is a special instance of
9 this situation.

- 1 **Exercise:** Derive the mean, variance, and standard error of the sample
2 mean. What is its MSE as an estimator of μ ?

$$E(\bar{X}) = \frac{1}{n} E(\sum X_i) = \frac{1}{n} (n\mu) = \mu$$

$\bar{X}$ is an unbiased estimator for μ

$$V(\bar{X}) = V\left(\frac{1}{n} \sum X_i\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

SE of $\bar{X}$ is $\sigma/\sqrt{n}$. Approx. using $s/\sqrt{n}$

$$\begin{aligned} \text{MSE of } \bar{X} &= V(\bar{X}) + \text{bias}^2(\bar{X}) = V(\bar{X}) + 0 \\ &= \sigma^2/n \end{aligned}$$

- 1 **Exercise:** If we did assume that $X_1, X_2, \dots, X_n$ are i.i.d., what could we say
2 about the distribution of the sample mean?

3 As long as n is large, the CLT says $\bar{X} \approx N(\mu, \sigma^2/n)$

4

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